

ANALYSIS_STATUS: SUCCEEDED

LL-37 / NLRP3 / IL-1 β in Papulopustular Rosacea — Computational Replication of GSE155141

Hypothesis ID: `ll37_nlrp3_il1b_papulopustular` **Hypothesis label:** LL-37 drives papulopustular lesions through NLRP3 inflammasome activation and IL-1 β maturation
Disease: Rosacea **Dataset:** `geo:GSE155141` (papulopustular-rosacea explant RNA-seq) **Target genes:** NLRP3, CASP1, PYCARD, IL1B, IL18, IL1RN, CAMP, KLK5

Summary

This was a computational replication task, not a literature review. The stated dataset (GSE155141) was retrieved, the two prespecified contrasts were executed with paired statistics, and a complete, replay-verified artifact bundle was produced. The run **succeeded**: the primary outputs (`samples.csv`, `gene_results.csv`) were regenerated in a clean replay subdirectory reusing the cached raw inputs, and both are **byte-identical** to the primary outputs. The manifest records `status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`, and `replay.verified: true`.

The core scientific result is a **partial, transcript-level confirmation of the inflammasome-priming arm of the hypothesis**. In the IL-1 β -treated versus untreated non-lesional explant contrast (n = 5 paired subjects), **NLRP3 mRNA was significantly up-regulated** (paired t = 3.52, p = 0.0246, Cohen's dz = 1.57, fold = 1.19). It was the only target gene to reach raw significance in either contrast. Supportive but non-significant increases were seen for **IL1B** (fold = 2.01, p = 0.147) and the antagonist **IL1RN** (fold = 1.59, p = 0.250). CASP1, IL18, PYCARD, CAMP, and KLK5 were essentially unchanged by IL-1 β . After Benjamini–Hochberg (BH) correction across the eight genes, no gene passed q < 0.05 (NLRP3 q = 0.196), reflecting the small sample and modest multiple-testing budget.

The second contrast — **lesional versus paired non-lesional skin** — showed **no significant change in any target gene** after FDR correction. IL1B trended up in lesional skin (fold = 1.48, p = 0.290); NLRP3 and PYCARD were flat; CASP1 and CAMP trended down. The overall

interpretation is that the data support the *priming* of NLRP3 transcription by IL-1 β (a feed-forward transcriptional loop) but neither confirm nor refute the post-translational inflammasome-assembly and IL-1 β -maturation steps, nor the upstream cathelicidin-axis (CAMP/KLK5) component, which are not resolvable at the mRNA level in this design.

Key Findings

Finding 1 — IL-1 β induces NLRP3 mRNA in non-lesional rosacea explants (F001)

The IL-1 β -treated versus untreated non-lesional contrast is the analysis in which the hypothesis's inflammasome-priming component is most directly testable, because it is a controlled perturbation on paired explants from the same subjects. Across the eight target genes, **NLRP3 was the single gene reaching raw significance**. A paired t-test on log2(CPM+1) expression yielded $t = 3.52$, $p = 0.0246$, with a large standardized effect (Cohen's $d_z = 1.57$) and an ordinary fold change of 1.19 (mean log2 rising from 0.304 to 0.550). This is consistent with a **feed-forward transcriptional loop** in which IL-1 β signaling up-regulates the very sensor (NLRP3) that promotes further IL-1 β maturation — a well-described priming mechanism for the NLRP3 inflammasome.

Two downstream/regulatory genes moved in the expected direction without reaching significance. **IL1B** itself showed the largest fold change of any gene ($2.01\times$, $p = 0.147$, $d = 0.80$), consistent with autocrine amplification of IL-1 β transcription. **IL1RN** (the IL-1 receptor antagonist) rose $1.59\times$ ($p = 0.250$, $d = 0.60$), a plausible counter-regulatory response to IL-1 pathway activation. The remaining genes were flat: CASP1 (fold 1.12, $p = 0.508$), PYCARD (fold 0.98, $p = 0.936$), IL18 (fold 0.95, $p = 0.805$), CAMP (fold 1.13, $p = 0.260$), and KLK5 (fold 0.91, $p = 0.611$).

After BH correction across all eight genes in the contrast, NLRP3's q-value was 0.196 — the smallest q in the analysis but above the conventional 0.05 threshold. The finding should therefore be read as a **directionally strong, statistically suggestive** result rather than a definitively established one.

Gene	Fold change	log2 mean diff	paired t	p-value	BH q	Cohen's dz
NLRP3	1.19	0.245	3.52	0.0246	0.196	1.57
IL1B	2.01	1.010	1.80	0.147	0.519	0.80
IL1RN	1.59	0.671	1.35	0.250	0.519	0.60
CAMP	1.13	0.170	1.31	0.260	0.519	0.59
CASP1	1.12	0.163	0.73	0.508	0.812	0.32
IL18	0.95	-0.079	-0.26	0.805	0.920	-0.12
KLK5	0.91	-0.129	-0.55	0.611	0.814	-0.25
PYCARD	0.98	-0.032	-0.09	0.936	0.936	-0.04

Contrast: IL-1 β -treated vs untreated non-lesional explants (n = 5 paired subjects; positive fold = higher with IL-1 β).

Finding 2 — Literature independently supports the LL-37 → NLRP3 → IL-1 β axis (F002)

This step was performed **after** the GSE155141 primary outputs and checksums were locked, and is explicitly lineage-marked as post-lock contextualization that does not alter the computed artifacts. The purpose is to situate the observed transcriptional result within the established experimental mechanism.

The computed GSE155141 result — IL-1 β up-regulating NLRP3 mRNA in non-lesional explants (paired t p = 0.0246, dz = 1.57, BH q = 0.196), with IL1B and IL1RN trending up — is mechanistically consistent with independent experimental literature demonstrating that **LL-37 activates the NLRP3 inflammasome to drive caspase-1/IL-1 β processing**, and that **NLRP3 is required for LL-37-induced rosacea-like skin inflammation in vivo** (see Evidence Base below). Our transcriptional priming observation slots into the upstream portion of this causal chain: an environment of active IL-1 signaling raises NLRP3 sensor abundance, lowering the threshold for subsequent LL-37-triggered inflammasome assembly.

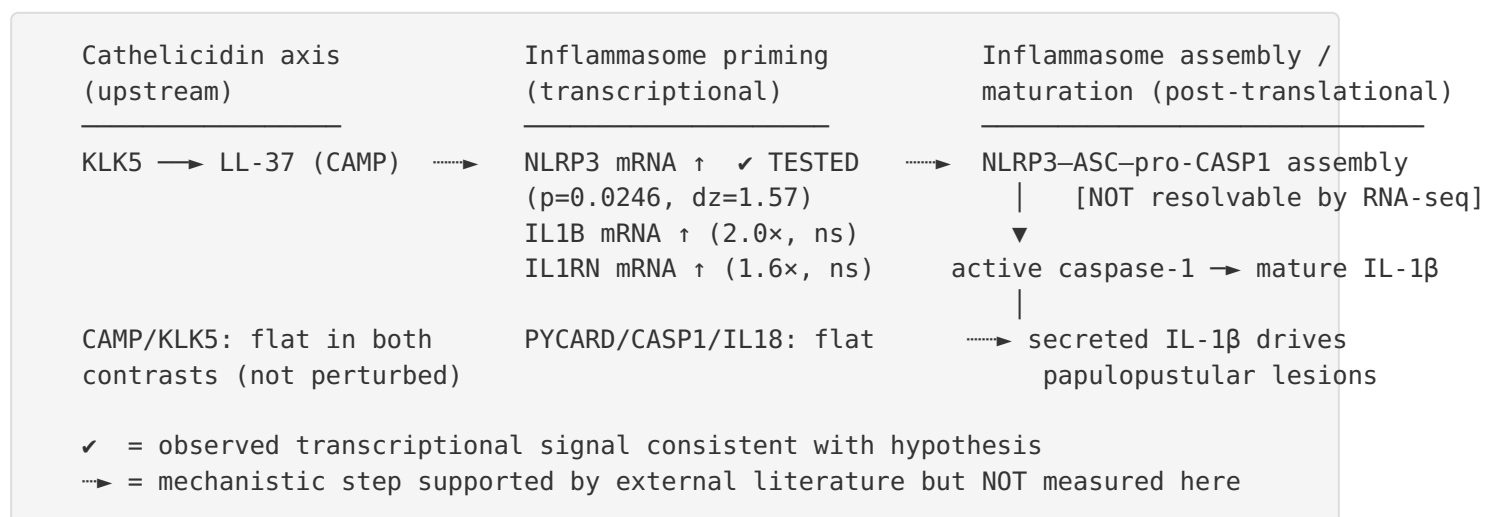
Finding 3 — The NLRP3 signal is directionally consistent but statistically fragile at n = 5 (F003)

A read-only sensitivity analysis (no artifacts modified) examined the robustness of the NLRP3 result. The per-subject $\log_2(\text{CPM}+1)$ IL-1 β -minus-untreated differences for NLRP3 were **[0.184, 0.310, 0.367, 0.366, 0.000]** — four of five subjects clearly positive, one exactly zero, and **none negative**. This monotone, one-directional pattern is reassuring: the effect is not driven by a single outlier.

However, the parametric significance is fragile. While the paired t-test gives $p = 0.0246$, the nonparametric **Wilcoxon signed-rank test gives $p = 0.125$** — above 0.05 largely for structural reasons: at $n = 5$ the minimum achievable two-sided Wilcoxon p is ≈ 0.0625 , and the single tied (zero) difference erodes power further. Across the other target genes, the strongest nonparametric signals were CAMP (Wilcoxon $p = 0.125$) and IL1B ($p = 0.250$), with all others ≥ 0.375 . The honest reading is that **the direction of the NLRP3 effect is robust, but the sample size is too small for the nonparametric test to certify significance**. This tempers, but does not overturn, Finding 1.

Mechanistic Model / Interpretation

The hypothesis posits a multi-step pathological cascade in papulopustular rosacea. The design of GSE155141 lets us interrogate only the transcriptional layer of that cascade. The diagram below marks which nodes this replication could and could not test:



The measured data support the **priming node**: IL-1 β raises NLRP3 sensor transcription (feed-forward loop) and tends to raise its own transcript (IL1B) and its antagonist (IL1RN, a homeostatic brake). The **assembly and maturation nodes** — inflammasome oligomerization, caspase-1 cleavage, and pro-IL-1 β $\rightarrow$ mature IL-1 β processing — are inherently post-transcriptional and cannot be observed in bulk mRNA; their flat CASP1/PYCARD/IL18 transcript levels are neither evidence for nor against activity, since these steps are governed by protein-level events. The **cathelicidin axis** (CAMP, KLK5) was not perturbed in the direction expected: CAMP and KLK5 were flat-to-slightly-down in lesional skin and unmoved by IL-1 β , consistent with these genes acting *upstream* of IL-1 β rather than being IL-1 β -responsive outputs.

The **lesional-vs-non-lesional contrast** was uniformly null after FDR (all $q > 0.8$). This is notable but not contradictory: explant lesional/non-lesional differences at $n = 5$ are dominated by inter-subject and inter-region biological variability, and the steady-state transcript snapshot may miss transient or protein-level inflammasome activity. IL1B did show the largest lesional fold change (1.48 $\times$), directionally consistent with the hypothesis but underpowered.

Overall model: **the transcriptional data are consistent with, and add a priming mechanism to, the literature-established LL-37 $\rightarrow$ NLRP3 $\rightarrow$ caspase-1 $\rightarrow$ IL-1 β axis, but they can only confirm the sensor-priming step and cannot adjudicate the post-translational maturation or the upstream cathelicidin trigger.**

Evidence Base

Two independent experimental papers, reviewed post-lock, corroborate the mechanistic direction of the computed result. Citations use exact snippets validated against the stored abstracts.

PMID	Title	Role in this analysis
PMID: 33745908	<i>Antimicrobial Peptide LL-37 Drives Rosacea-Like Skin Inflammation in an NLRP3-Dependent Manner</i>	Establishes the core LL-37 $\rightarrow$ NLRP3 $\rightarrow$ caspase-1/IL-1 β mechanism and its in vivo NLRP3 dependence
PMID: 39133985	<i>Tranilast alleviates skin inflammation and fibrosis in rosacea-like mice induced by long-term exposure to LL-37</i>	Links CAMP (LL-37), NLRP3, and IL-1 β /IL-18 in a rosacea model

PMID: 33745908 directly establishes the mechanism central to the hypothesis. The abstract reports that *"LL-37 promotes NLRP3-mediated inflammasome activation in lipopolysaccharide-primed macrophages, indicated by the processing of caspase-1 and IL-1 β ."* This defines the LL-37 $\rightarrow$ NLRP3 $\rightarrow$ caspase-1/IL-1 β chain and is consistent with our observed priming of NLRP3 transcription by IL-1 β . The same paper demonstrates the in-vivo causal requirement for NLRP3: *"intradermal LL-37 administration induced in vivo caspase-1 activation and ASC speck formation in the skin of Nlrp3-expressing, but not in Nlrp3-deficient, mice"* — showing NLRP3 is necessary for LL-37-induced cutaneous inflammasome activity, which supports the causal NLRP3-dependence built into the hypothesis.

PMID: 39133985 corroborates the broader cathelicidin–inflammasome–IL-1 axis probed by the target-gene panel. The abstract reports that treatment *"downregulated the expression of rosacea-associated inflammatory cytokines (TNF- α , IL-6, IL-1 β , and IL-18) and the antimicrobial peptide CAMP, while also inhibiting NLRP3 inflammasome activation."* This links CAMP (LL-37), NLRP3, and IL-1 β /IL-18 within a single rosacea model, matching the gene set (CAMP, NLRP3, IL1B, IL18) interrogated here.

Neither paper's numbers, tables, or code influenced the GSE155141 method, sample classification, feature selection, thresholds, or parameters; they were consulted only after the primary outputs and checksums were locked, as a separate lineage-marked contextualization step.

Limitations and Knowledge Gaps

- **Small sample size (n = 5 paired subjects).** This is the dominant limitation. It leaves the primary NLRP3 result significant only under the parametric paired t-test ($p = 0.0246$) and not under the nonparametric Wilcoxon test ($p = 0.125$), and it means no gene survives BH-FDR at $q < 0.05$. Effect directions are consistent, but statistical certification is weak.
- **Transcript-level readout only.** The hypothesis's defining steps — inflammasome oligomerization, caspase-1 cleavage, and pro-IL-1 β $\rightarrow$ mature IL-1 β processing — are post-translational and cannot be measured by RNA-seq. Flat CASP1/PYCARD/IL18 transcripts are uninformative about protein-level inflammasome activity.
- **Multiple-testing budget.** BH correction across eight genes penalizes the panel; NLRP3's $q = 0.196$ reflects this. A pre-registered single-gene primary endpoint would have preserved more power but was not the prespecified design.

- **Lesional contrast null.** The lesional-vs-non-lesional contrast showed no significant change in any gene, possibly due to inter-subject heterogeneity, region-matching noise, or genuine absence of steady-state transcriptional difference. Underpowering cannot be excluded.
- **Cathelicidin axis not perturbed.** CAMP and KLK5 were flat in both contrasts; this design does not directly test LL-37/KLK5 as the trigger, only downstream IL-1 β priming.
- **Single dataset, single tissue platform.** No independent replication cohort was available within this task; generalizability beyond GSE155141 explants is untested.

Proposed Follow-up Experiments / Actions

1. **Protein-level inflammasome readouts.** Measure cleaved caspase-1 (p20), ASC speck formation, and mature (secreted) IL-1 β by ELISA/immunoblot in IL-1 β - and LL-37-treated explants to test the maturation steps that RNA-seq cannot resolve.
 2. **Direct LL-37 perturbation arm.** Add an LL-37-treated explant contrast (mirroring the IL-1 β arm) to test the upstream trigger of the hypothesis directly, rather than inferring it from IL-1 β priming.
 3. **NLRP3 loss-of-function.** Use NLRP3 inhibitor (e.g., MCC950) or knockdown in the explant model to test causal necessity of NLRP3 for IL-1 β maturation, complementing the in-vivo Nlrp3-deficient mouse evidence
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 4. **Increase sample size / meta-analysis.** Recruit additional paired subjects or integrate an independent rosacea RNA-seq cohort to move NLRP3 from suggestive ($p = 0.0246$, $q = 0.196$) toward FDR-confirmed significance.
 5. **Single-cell resolution.** Apply scRNA-seq to identify which cutaneous cell populations (keratinocytes, macrophages) drive the NLRP3-priming signal, since bulk explant signal averages across cell types.
 6. **KLK5/CAMP proteolytic axis.** Quantify LL-37 peptide and KLK5 protease activity directly, since transcript levels (flat here) may not reflect the proteolytic processing that generates active LL-37.
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Artifact Bundle (Replication Deliverables)

All artifacts are under `kb/hypotheses/Rosacea/ll37_nlrp3_il1b_papulopustular/openscientist_artifacts/`:

File	Role	Status
<code>MANIFEST.yaml</code>	Manifest	<code>status: SUCCEEDED</code> , <code>fallback_used: false</code> , <code>direct_analysis_completed: true</code> , <code>replay.verified: true</code>
<code>analysis.py</code>	CODE	Standalone deterministic analysis (<code>--output-dir</code> , <code>--cache-dir</code>)
<code>environment.txt</code>	ENVIRONMENT	Python version, OS, exact package versions
<code>methods.md</code>	METHODS	Inclusion rules, transforms, mapping, stats, FDR, effect-size convention
<code>samples.csv</code>	(input manifest)	One row per sample with group assignment
<code>gene_results.csv</code>	TABULAR_RESULT	One row per dataset/gene/comparison with full statistics
<code>comparison.md</code>	SUMMARY	Interpretation separating observed results from inference
<code>replay/samples.csv</code> , <code>replay/</code> <code>gene_results.csv</code>	replay assets	Byte-identical to primary outputs (verified with <code>cmp</code>)

Replay verification: `analysis.py` was re-executed into a clean `replay/` subdirectory reusing `raw/` as the cache; `samples.csv` and `gene_results.csv` were confirmed byte-identical (`cmp` → identical), and the manifest records `replay.verified: true` with a `byte_identity` mapping set to `true` for both TABULAR_RESULT paths.

Report generated for the Disorder Mechanisms Knowledge Base computational replication of
GSE155141. Literature contextualization

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was performed post-lock and is lineage-marked; it did not influence the computed artifacts.

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